OpenDRIVE® 1.8.0, Static Road Network Description, 2023-11-22

7 Junctions with entry and exit lanes

7.1 Introduction

At a junction with entry or exit lanes a driver can join or leave a road on a dedicated lane for acceleration or deceleration.

Junctions with entry and exit lanes are typical for:

- Motorways
- Rural junctions

This example is about motorways.

There are two different types of motorway entries and exits defined in ASAM OpenDRIVE:

- Non-overlapping
- Overlapping

The following sections describe the modeling approaches for each type.

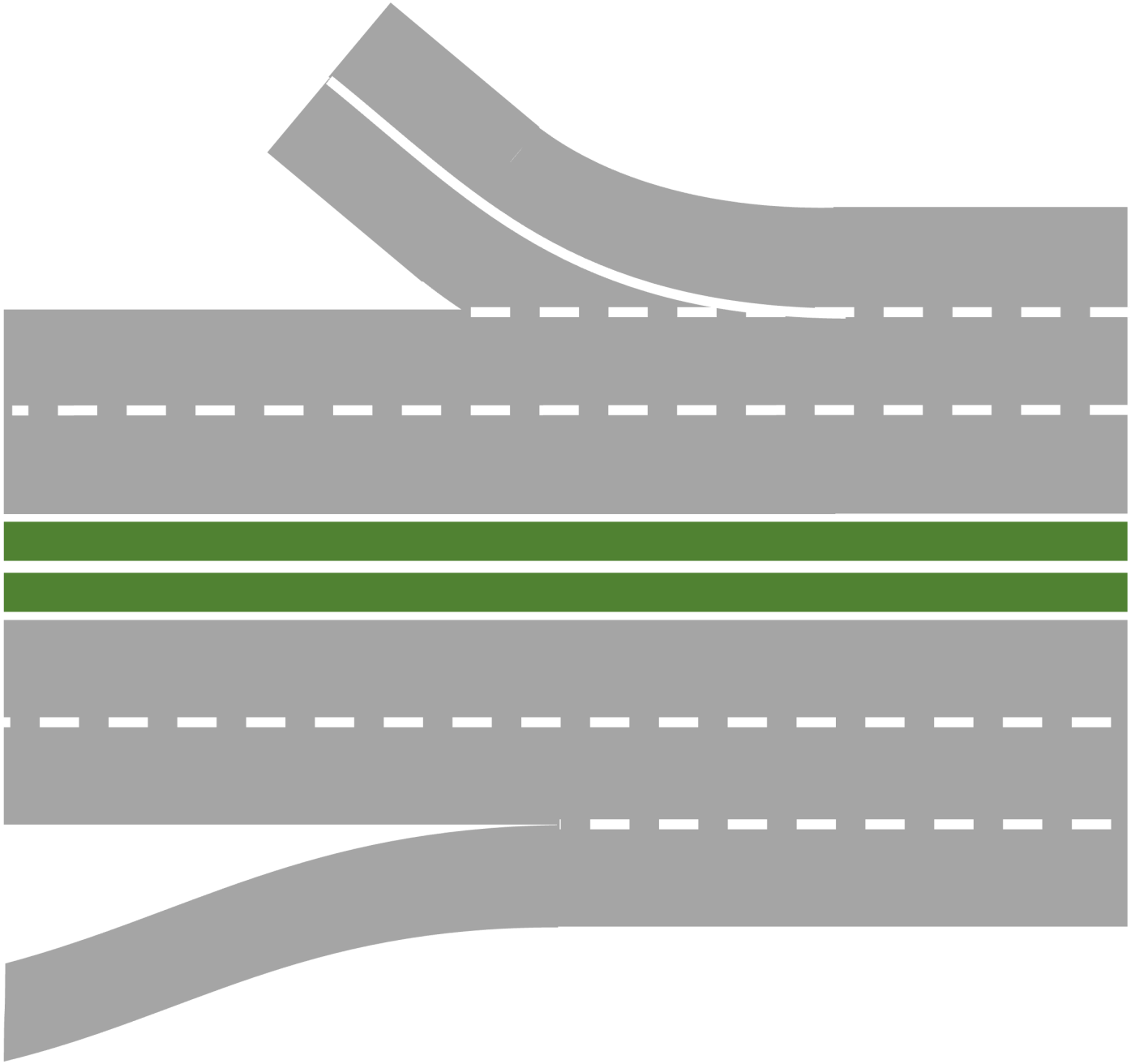

Figure 33. Motorway with entry lanes and exit lanes

Figure 33 shows a non-overlapping entry and an overlapping exit at a motorway.

7.1.1 Rules and guidelines

The recommendation is to use a direct junction to connect an exit lane or an entry lane. The direct junction allows one overlapping lane. The road with the entry lane or exit lane and the main road can overlap where they split or merge.

- If traffic of different lanes is crossing at the junction, a common junction shall be used.
- The overlap zone shall be as small as possible.
- Direct junctions require one *cut*, common junctions require two *cuts*. Therefore direct junctions are recommended, if possible.
- The rules for road linkage apply, see Section 10.3, "Road linkage".
- The rules for lane types apply, see Section 11.7, "Additional lane properties".

7.1.2 Challenges

- Choose the optimal junction type.
- Select the location for the junction.
- Assign the correct lane types.

7.2 Non-overlapping

A non-overlapping entry or exit is a designated lane that does not overlap with another lane. It has only one drivable direction.

Non-overlapping entries or exits can be modeled with direct junctions or common junctions. The recommendation is to use a direct junction to connect an entry or exit with the main road.

7.2.1 Example

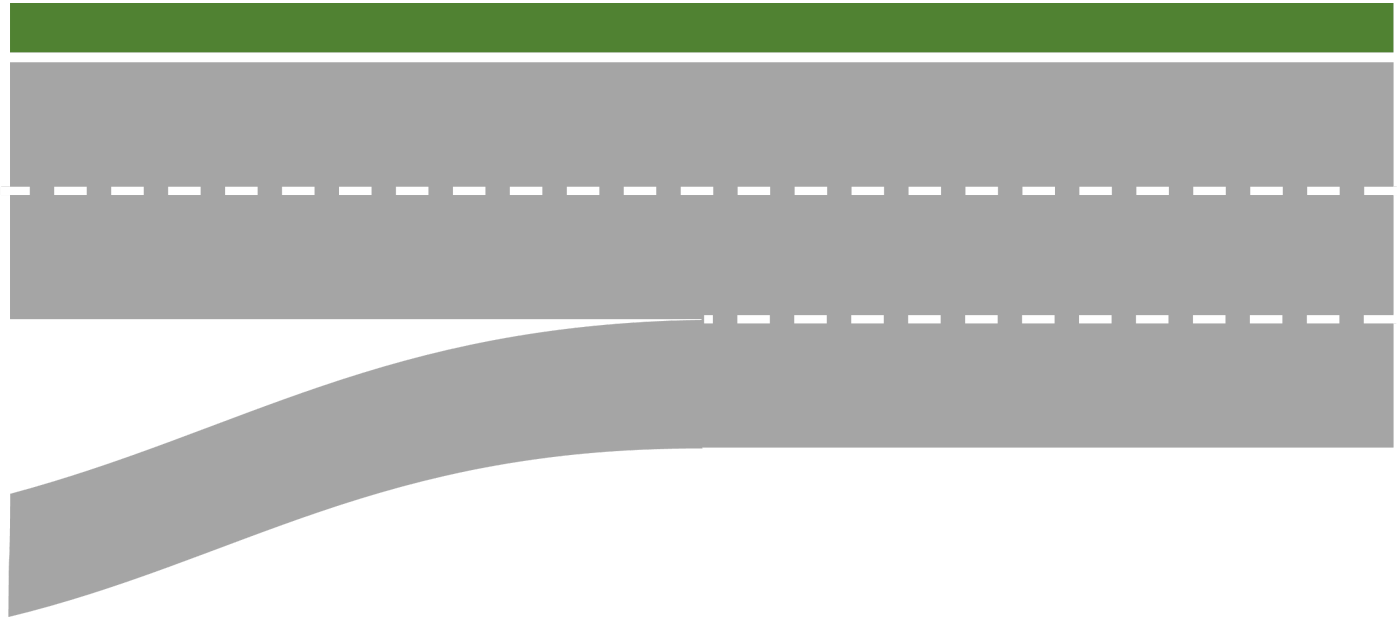

Figure 34. Motorway with an entry

Figure 34 shows a motorway with a non-overlapping entry.

7.2.2 Modeling approach

7.2.2.1 Junction type

Direct junctions are the best choice when roads split or merge.

1. Connect the roads with a direct junction at the position where the the entry joins the main road.

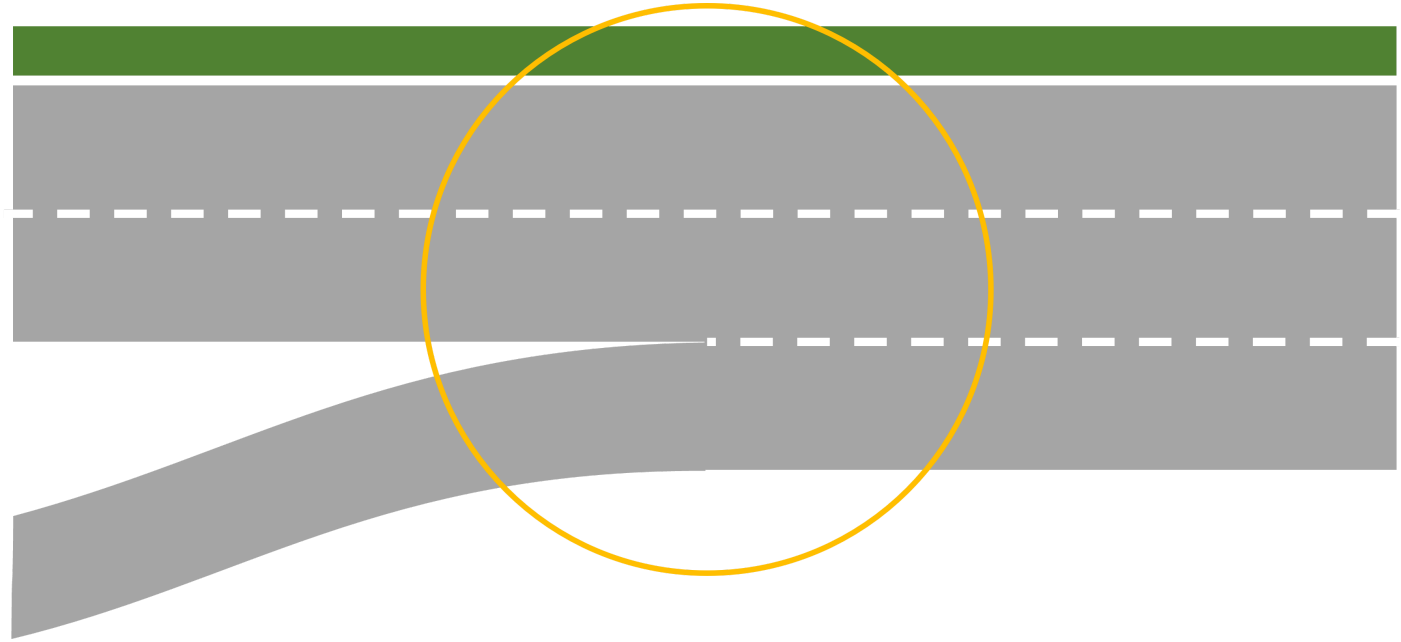

Figure 35. Direct junction

Figure 35 shows the area with the direct junction marked with a circle.

7.2.2.2 Road reference lines

1. Place the road reference lines for the roads to start or end at the direct junction.

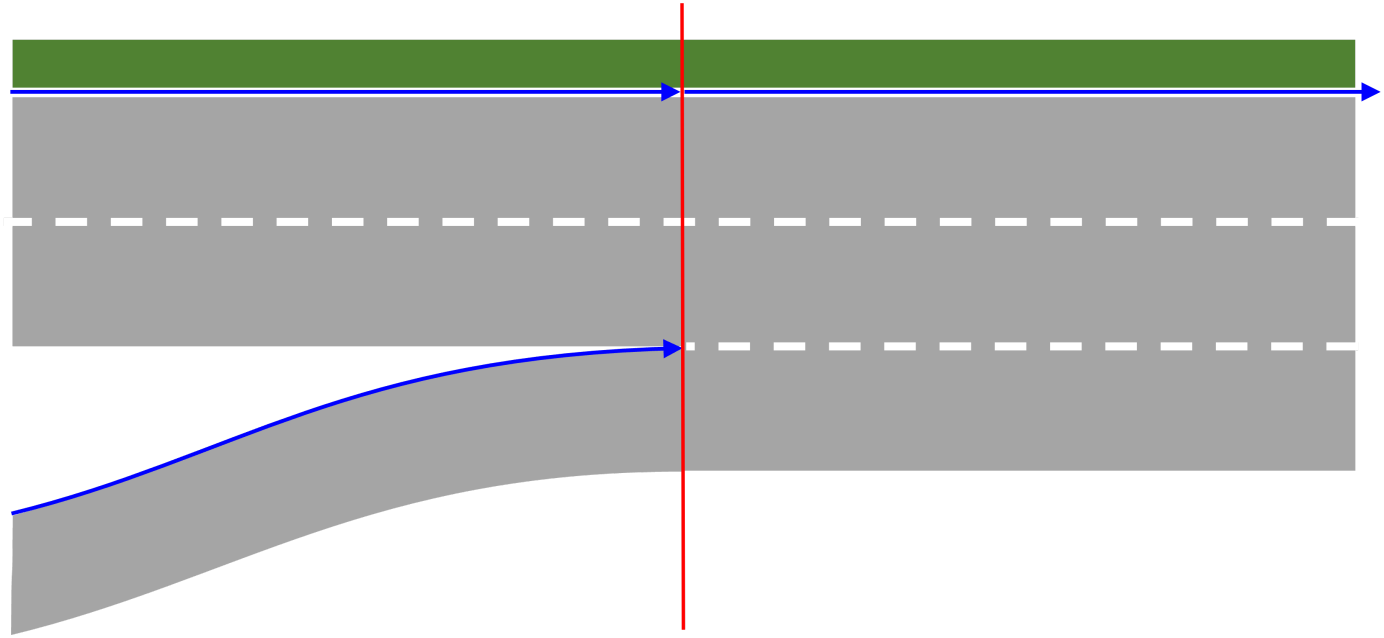

Figure 36. Road reference lines

Figure 36 shows the road reference lines (blue arrows) and the direct junction (red line).

7.2.2.3 Lane types

1. Set the value of the @type attribute for the lane on the joining road before the direct junction to `onRamp`, the value of the @type attribute of the outermost lane after the direct junction to `entry`, and the value of the @type attribute of the other lanes to `driving`.

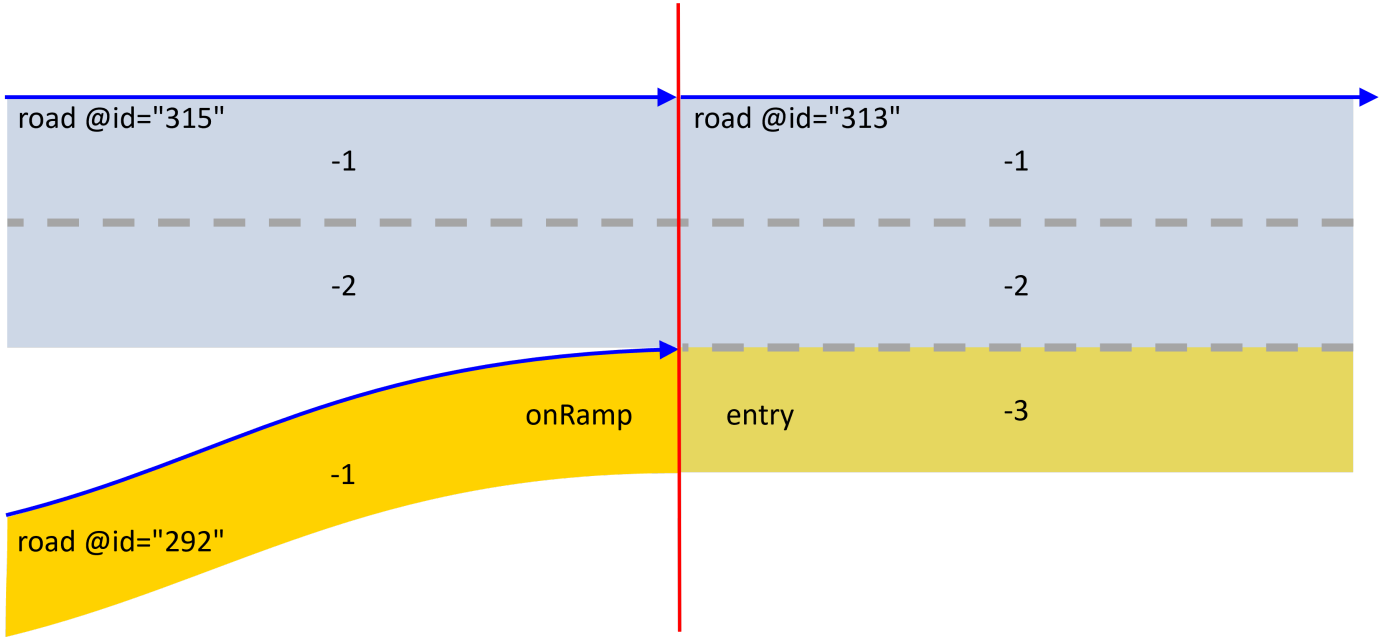

Figure 37. Motorway with a non-overlapping entry

Figure 37 shows a motorway with an entry:

- The road with @id="315" has the two lanes with @id="-1" and @id="-2", both with @type="driving".
- Next to the road with @id="315", there is the road with @id="292". The road with @id="292" has one lane with @id="-1" and @type="onRamp".
- The road with @id="315" and the road with @id="292" connect with a direct junction to the road with @id="313".
- The direct junction is at the gore point, where the road with @id="292" and the road with @id="315" merge into the road with @id="313".
- The road with @id="313" has three lanes. Two of them, the lanes with @id="-1" and @id="-2", are lanes with @type="driving" and one of them, the lane with @id="-3", is a lane with @type="entry".
- The lane with @id="-1" of the road with @id="292" and the lane with @id="-2" of the road with @id="315" do not overlap.
- The lane with @id="-1" of the road with @id="292" and the lane with @id="-3" of the road with @id="313" do not overlap with other lanes.

The motorway entry is non-overlapping and traffic does not cross. Therefore a direct junction is possible.

XML example

Solution in ASAM OpenDRIVE 1.8.0

- Ex_Entry_Exit.xodr

7.3 Overlapping

An overlapping entry or exit is a lane that overlaps with another lane. It has only one drivable direction. Overlapping entries and exits shall be modeled with direct junctions if possible. If more than one lane overlaps or traffic crosses, a common junction shall be used.

7.3.1 Example

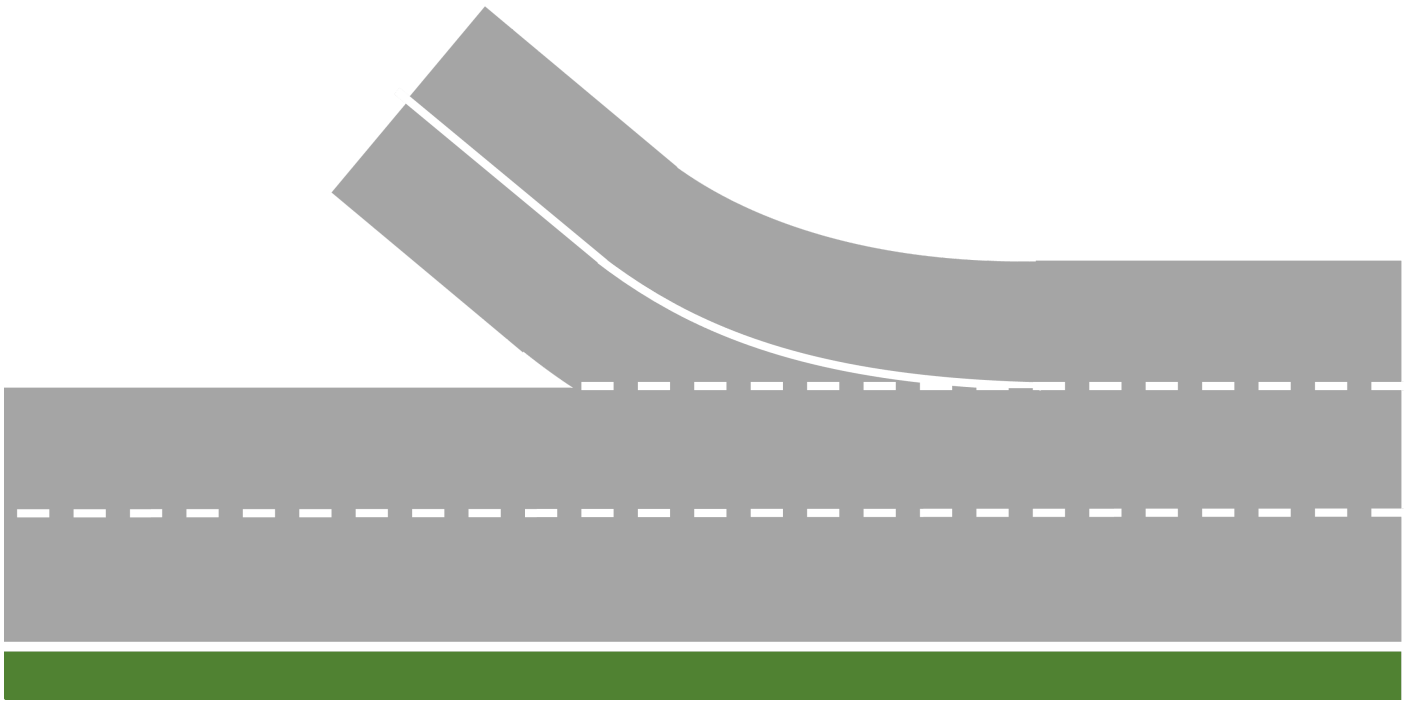

Figure 38. Motorway with an exit

Figure 38 shows a motorway with an overlapping exit.

7.3.2 Modeling approach

7.3.2.1 Junction type

Direct junctions are the best choice when roads split or merge. Direct junctions allow one lane to overlap.

1. Connect the roads with a direct junction at the position where the exit branches from the main road.

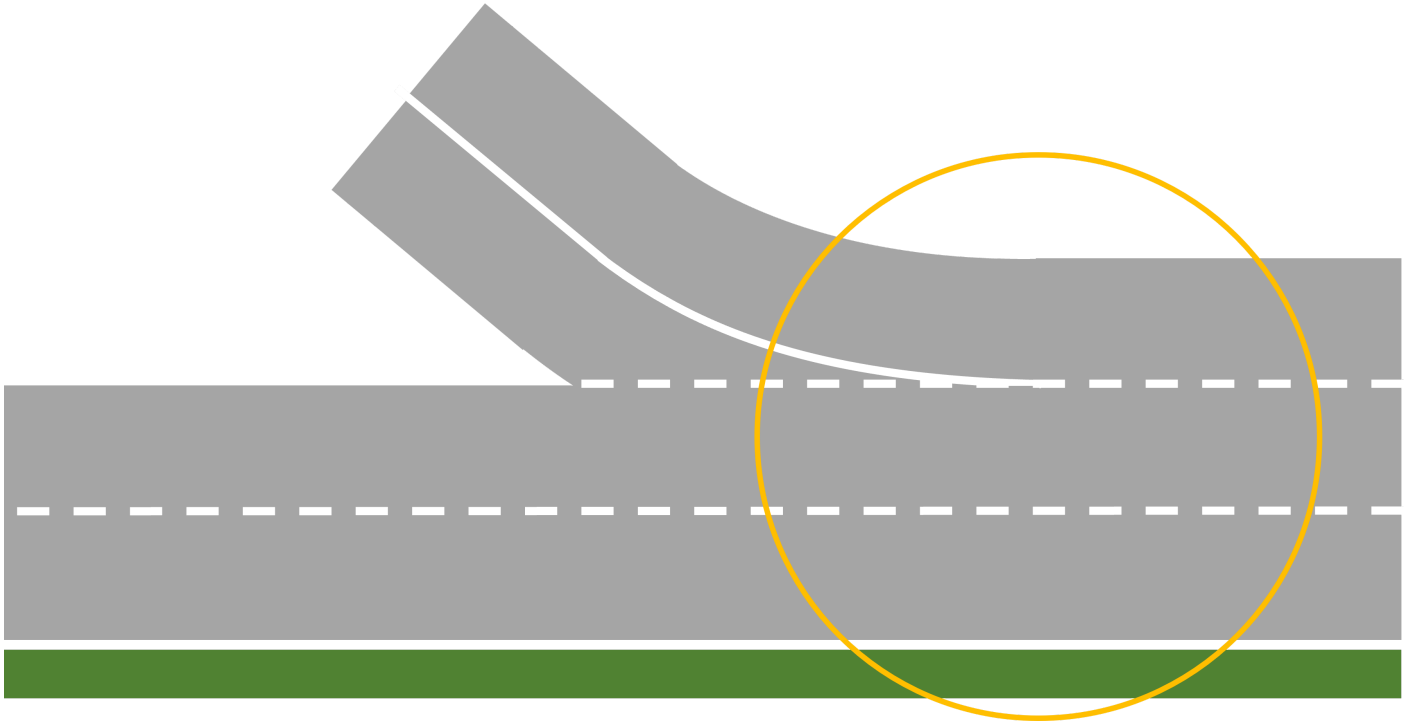

Figure 39. Direct junction

Figure 39 shows the area with the direct junction marked with a circle.

7.3.2.2 Road reference lines

1. Place the road reference lines for the roads to start or end at the direct junction.

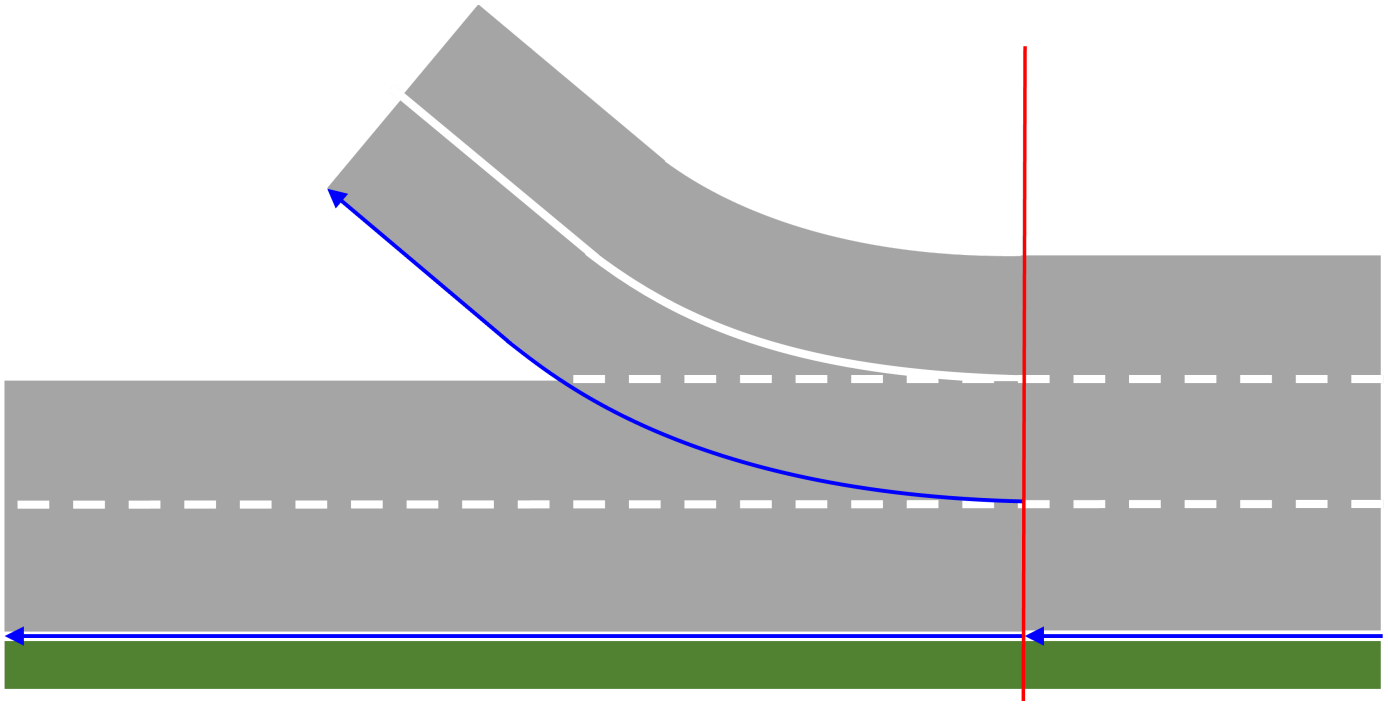

Figure 40. Road reference lines

Figure 40 shows the road reference lines (blue arrows) and the direct junction (red line).

7.3.2.3 Lane types

1. Set the value of the @type attribute for the outermost lane before the direct junction to `exit`. After the direct junction set the value of the @type attribute for the outermost lane at the exiting road to `offRamp`, and set the value of the @type attribute for the overlapping lane to `exit`.
2. Add a lane section to the exiting road after the overlapping area to change the value of the @type attribute from `exit` to `offRamp`.
3. Apply the usual values to the @type attributes of the other lanes, for example `driving`.

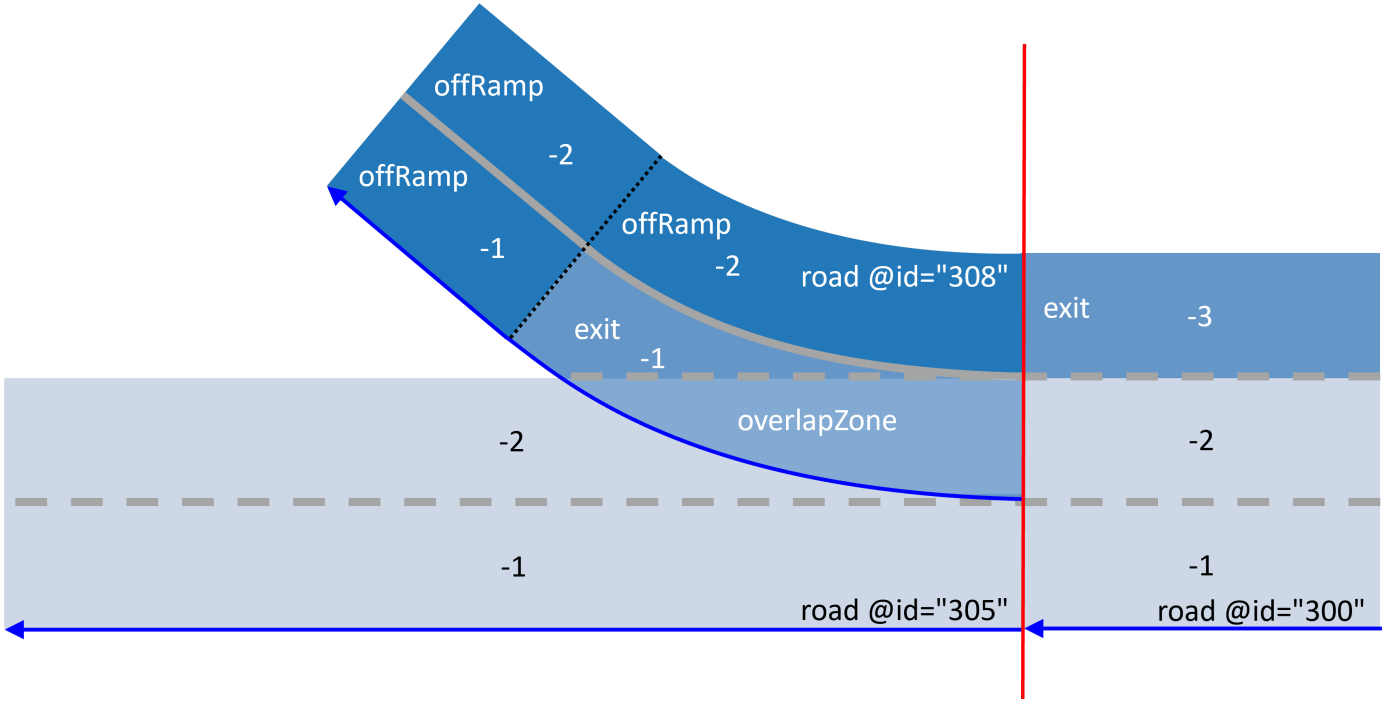

Figure 41. Motorway with an overlapping exit

Figure 41 shows a motorway with an exit:

- The road with @id="300" has three lanes. Two of them, the lanes with @id="-1" and @id="-2", are lanes with @type="driving" and one of them, the lane with @id="-3", is a lane with @type="exit".
- The road with @id="300" connects with a direct junction to two roads, the road with @id="305" and the road with @id="308".
- The direct junction is at the gore point, where the road with @id="300" splits into the road with @id="305" and the road with @id="308".
- The road with @id="305" has two lanes, the lanes with @id="-1" and @id="-2", both with @type="driving".
- The road with @id="308" has two lanes, the lane with @id="-1" and @type="exit" and the lane with @id="-2" and @type="offRamp".
- The lane with @id="-2" of the road with @id="300" splits into two overlapping lanes, the lane with @id="-2" and @type="driving" of the road with @id="305" and the lane with @id="-1" and @type="exit" of the road @id="308".

The motorway exit is overlapping, but only one lane overlaps and traffic does not cross. Therefore a direct junction is possible.

- The road with @id="308" has two lane sections, one lane section from the direct junction to the end of the overlapping area and a second lane section thereafter.
- After the overlapping area, the road with @id="305" continues unchanged. The lane with @id="-1" of the road with @id="308" changes the value of the @type attribute from `exit` in the first lane section to `offRamp` in the second lane section. The lane with @id="-2" of the road with @id="308" continues unchanged.

7.3.2.4 Overlap zone

1. Set the @overlapZone attribute at the lane links of the two overlapping lanes.

The overlap zone has to be at least as long as the overlapping area but as small as possible.

XML example

Solution in ASAM OpenDRIVE 1.8.0

- Ex_Entry_Exit.xodr

7.4 Related topics

- Section 10.3, "Road linkage"
- Section 11.7.1, "Lane type"
- Section 12.2, "Common junctions"
- Section 12.6, "Direct junctions"